

**Replication of a Research Claim from Axt et al., (2018), from the
Journal of Experimental Social Psychology**

Replication Team: Victoria Colvin and Colin Tucker Smith
University of Florida, Gainesville, FL

Research Scientist: Nick Fox
Action Editor: Richard Lucas

Project ID: Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith - Direct Replication - m5g9

OSF project: <https://osf.io/etfqz/>

Preregistration: <https://osf.io/wg5ny>

https://docs.google.com/document/d/1BzaD4xWCG1JM-2XOQLdSCL16PMvRFXLp3l_p6bNzw1A/edit?usp=drivesdk

Replication of a research claim from Axt et al., (2018), from the Journal of Experimental Social Psychology

Claim Summary

This replication tests the claim from Study 1 of Axt et al., (2018) that members of a nondominant racial group will show reliable implicit ingroup preference when presented with positively valenced information, and will show reliable implicit preference for the dominant group when presented with negative valenced information. We calculated D scores from the good focal Multiple Category – Implicit Association Test (MC-IAT) and bad MC-IAT output using the D algorithm (Nosek et al., 2014). To test the central claim, we conducted an independent samples t -test to determine whether there was a significant difference in implicit ingroup preference between the good focal and bad focal conditions. We also performed one sample t -tests for both the bad focal and good focal conditions to test if preference within condition differed from zero. Additionally, explicit preferences were tested against 0 (i.e., no preference) using one-sample t -tests. We also tested the relationship between implicit and explicit measures using correlations. Finally, we used Fisher's r to z transformation to compare independent correlations in testing whether explicit preferences are more related to the good-focal MCIAT or the bad-focal MCIAT.

Replication Criteria. A successful replication of the Axt et al., (2018) results would be a statistically significant effect ($\alpha = .05$, two tailed) in the same pattern as the original study. For the replication of the central claim, this criteria is met by a statistically significant difference between good-focal MC-IAT and bad-focal MC-IAT conditions, with bad-focal having a higher (more positive) D score.

Replication Result. The replication included 372 participants in the analytic sample to reach the requirements for Stage 1 data collection. An independent samples t -test revealed that there is a

significant difference in implicit ingroup preference between the good-focal MC-IAT D score ($M = -0.25$, $SD = 0.47$) and bad-focal MC-IATs D score ($M = 0.01$, $SD = 0.56$), $t(370) = 4.97$, $p = .0000010255$, $d = 0.51$. Thus, the replication of the central claim was successful according to the SCORE criteria. The replication attempt was powered to test only the central claim, as such we only report the results of the other statistical tests and do not interpret them.

Specifically, we computed one-sample t -tests to assess whether implicit preferences differed from zero (i.e., no preference). In the bad-focal condition, implicit preference did not differ from zero, $t(175) = 0.28$, $p = .782$, $d = 0.02$. In contrast, implicit preference did differ from zero in the good-focal condition, $t(195) = -7.56$, $p = .001$, $d = 0.53$. To determine whether explicit beliefs ($M = 1.28$, $SD = -1.23$) significantly differed from zero we again ran a one-sample t -test, $t(370) = -20.10$, $p = .001$, $d = 1.04$. Correlations between the explicit preference measures, bad-focal MC-IAT, and good-focal MC-IAT, as well as means and standard deviations for each measure are in the Appendix. Finally, the Fisher r to z transformation was conducted to test whether explicit preferences were more related to the good-focal MC-IAT or bad-focal MC-IAT. Using a two-tailed test of significance, the correlation between explicit preference and good-focal MC-IAT ($r = 0.13$, $p = .06$) did not differ from the correlation between explicit preference and bad-focal MC-IAT, ($r = 0.21$, $p = .004$), $z = -0.79$, $p = .429$.

Methods & Materials

The following materials are publicly available on the OSF site:

- A **materials** file that describes the components of the survey, with a link to a copy of the Project Implicit coding, provided as a WORD document and a PDF file. Filenames:
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-materials.docx
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-materials.pdf
- The **raw data** files downloaded from Project Implicit are provided in two folders. One folder contains the .txt version of all the raw data files, and the other folder contains the .csv version of all the raw data files. Folder names:
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9- RAW DATA.csv
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-RAW DATA.txt
- The **analytic data** file saved as a .csv file and as an .sav file. Filenames:
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-analytic dataset.csv
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-analytic dataset.sas
- A **data dictionary** for every variable in the analytic dataset, provided as a pdf file.
Filename:

- SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-data dictionary for analytic dataset.pdf
- The **data analysis script**, provided as a WORD doc. This script transforms the raw data into the analysis data set and executes the analyses to get to the final results. Filename:
 - SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-script for producing key hypothesis test from raw data.docx
- The **output** from the SAS analyses, available in a pdf file. Filename:
 - SCORE - Axt_JournExpSocPsych_2018_zk2 - Colvin_Smith-m5g9-Output.pdf

The entire data transformation and analysis can be replicated by running the data analysis script, pasted into the SAS Syntax window (SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-script for producing key hypothesis test from raw data.docx) on the raw data files (SCORE-Axt_JournExpSocPsych_2018_zK2 - Colvin_Smith- m5g9-RAW DATA.txt).

Deviations from the Preregistration

In the preregistration we specify that only Black U.S. citizens would be allowed to take the study. Upon deployment of the study, a member of the Scientific Advisory Committee for Project Implicit expressed concerns that limiting the study to only Black participants would negatively impact the collection of Black participants for other studies in the research pool, therefore the study requirements were adjusted to allow U.S. citizens of any race to take the study. Importantly, only data from Black participants is analyzed for the purposes of this replication.

References

- Axt, J. R., Moran, T., & Bar-Anan, Y. (2018). Simultaneous ingroup and outgroup favoritism in implicit social cognition. *Journal of Experimental Social Psychology*, 79, 275–289. <https://doi.org/10.1016/j.jesp.2018.08.007>
- Nosek, B. A., Bar-Anan, Y., Sriram, N., Axt, J., & Greenwald, A. G. (2014). Understanding and using the Brief Implicit Association Test: Recommended scoring procedures. *PLoS ONE*, 9, e110938. <https://doi.org/10.1371/journal.pone.0110938>

Appendix

Table 1

Correlation matrix for implicit and explicit attitudes

Measure	1	2	3
1. Explicit beliefs	-		
2. Good MC-IAT	.13 ⁺	-	
3. Bad MC-IAT	.21**	-	-

Note: ⁺ $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$

Table 2

Means and standard deviations for implicit and explicit attitudes

Measure	<i>N</i>	<i>M(SD)</i>
Good MC-IAT	176	-0.25 (0.47)
Bad MC-IAT	196	0.01 (0.56)
Explicit beliefs	371	-1.28 (1.23)